

n-type thermoelectric material $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ for high power generation

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Contributed by Ching-Wu Chu, December 22, 2014 (sent for review September 3, 2014; reviewed by Renkun Chen and Kornelius Nielsch)

Thermoelectric power generation is one of the most promising techniques to use the huge amount of waste heat and solar energy. Traditionally, high thermoelectric figure-of-merit, ZT , has been the only parameter pursued for high conversion efficiency. Here, we emphasize that a high power factor (PF) is equivalently important for high power generation, in addition to high efficiency. A new n-type Mg_2Sn -based material, $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$, is a good example to meet the dual requirements in efficiency and output power. It was found that $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ has an average ZT of 0.9 and PF of $52 \mu\text{W}\cdot\text{cm}^{-1}\cdot\text{K}^{-2}$ over the temperature range of 25–450 °C, a peak ZT of 1.4 at 450 °C, and peak PF of $55 \mu\text{W}\cdot\text{cm}^{-1}\cdot\text{K}^{-2}$ at 350 °C. By using the energy balance of one-dimensional heat flow equation, leg efficiency and output power were calculated with $T_h = 400$ °C and $T_c = 50$ °C to be of 10.5% and $6.6 \text{ W}\cdot\text{cm}^{-2}$ under a temperature gradient of 150 °C $\cdot\text{mm}^{-1}$, respectively.

thermoelectrics | magnesium | tin | power factor | output power

Thermoelectric power generation from waste heat is attracting more and more attention. Potential fuel efficiency enhancement by recovering the waste heat is beneficial for automobiles and many other applications (1, 2). In addition, solar thermoelectric generator provides an alternative route to convert solar energy into electrical power besides the photovoltaic conversion (3). Thermoelectric generator (TEG) can be regarded as a heat engine using electrons/holes as the energy carrier. The conversion efficiency of a TEG is related to the Carnot efficiency and the material's average thermoelectric figure of merit ZT (4):

$$\eta = \frac{T_h - T_c}{T_h} \left(\frac{\sqrt{1 + ZT} - 1}{\sqrt{1 + ZT} + T_c/T_h} \right), \quad [1]$$

where $ZT = (S^2\sigma/\kappa)T$, and S , σ , κ , and T are Seebeck coefficient, electrical conductivity, thermal conductivity, and absolute temperature, respectively. Pursuing high ZT has been the focus of the entire thermoelectric community by applying various phonon engineering via nanostructuring approaches to reduce the thermal conductivity (5–7), or by exploring new compounds with intrinsically low thermal conductivity, such as compounds having complex crystalline structure, local rattlers, liquid-like sublattice, and highly distorted lattice (8–11). However, for practical applications, efficiency is not the only concern, and high output power density is as important as efficiency when the capacity of the heat source is huge (such as solar heat), or the cost of the heat source is not a big factor (such as waste heat from automobiles, steel industry, etc.). The output power density ω is defined as the output power W divided by the cross-sectional area A of the leg, i.e., $\omega = W/A$, which is related to power factor $PF = S^2\sigma$ by the following:

$$\omega = \frac{1}{4} \frac{(T_h - T_c)^2}{L} PF. \quad [2]$$

Eq. 2 contains two main parts: square of the temperature difference divided by leg length, and material power factor $PF = S^2\sigma$.

Clearly, to achieve higher power density for a given heat source, we have to either increase the power factor PF or decrease the leg length. However, decreasing the leg length could cause severe consequences such as increase of large heat flux that will increase the cost of the heat management at the cold end, increase of percentage of contact resistance in the device that will increase the parasitic loss and consequently decrease the energy conversion efficiency, increase of the thermal stress due to the larger thermal gradient leading to device failure, etc. Therefore, it is better to increase the power factor PF . Because PF is a pure material parameter, we can use it as a criterion in searching for new thermoelectric materials for high output power.

A useful thermoelectric material should possess high ZT value for high efficiency, and also very importantly high PF for high output power. Ideally, temperature-independent ZT and PF over the whole temperature range from cold side to hot side are desired. However, both the ZT and PF of all materials show strong temperature dependency, usually increasing first with temperature and then decreasing when bipolar effect starts to play a role. The working temperature of thermoelectric materials is limited by the band energy gap E_g ; e.g., Bi_2Te_3 , a well-known thermoelectric material for applications below 200 °C, has an E_g of ~ 0.13 eV (12). PbTe and associated materials have much higher peak ZT in the temperature range of 400–600 °C due to its larger E_g of 0.32 eV (13). However, the toxicity of lead, poor mechanical properties, and thermal instability above 400 °C seriously limit the application of Pb-based thermoelectric materials. Even though Mg_2Si , skutterudites, and half-Heuslers are promising

Significance

Thermoelectric materials have been extensively studied for applications in conversion of waste heat into electricity. The efficiency is related to the figure-of-merit, $ZT = (S^2\sigma/\kappa)T$, where S , σ , and κ are the Seebeck coefficient, electrical conductivity, and thermal conductivity, respectively. Pursuing higher ZT for higher efficiency has been the focus by mainly reducing the thermal conductivity. In this paper, we point out, for a given ZT , higher power factor ($S^2\sigma$) should be pursued for achieving more power because power is determined by $(T_h - T_c)^2(S^2\sigma)/L$, where T_h , T_c , and L are the hot and cold side temperatures, and leg length, respectively. We found a new material, $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$, having both high ZT and high power factor.

Author contributions: W.L. and Z.F.R. designed research; W.L., Q.J., and B.L. performed research; W.L., H.S.K., S.C., Q.J., B.L., M.Y., and Z.S.R. contributed new reagents/analytic tools; W.L., C.P.O., S.W., C.-W.C., and Z.F.R. analyzed data; and W.L., H.S.K., C.-W.C., and Z.F.R. wrote the paper.

Reviewers: R.C., University of California, San Diego; and K.N., University of Hamburg.

The authors declare no conflict of interest.

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This article contains supporting information online at www.pnas.org/lookup/suppl/doi:10.1073/pnas.1424388112/-DCSupplemental.

for thermoelectric power generation at up to 500 °C [Mg_2Si (14) and skutterudites (15, 16)] or 600–700 °C [half-Heuslers (17, 18)], the ZT values of these materials below 400 °C is relatively low ($ZT < 1$). Other materials, such as n-type $\text{In}_4\text{Se}_{3.8}$ (19), n-type $\text{Ba}_8\text{Ga}_{16}\text{Sn}_{30}$ (20), and p-type $\text{Zn}_{4.8}\text{Sb}_3$ (21) have higher average ZT values below 400 °C. However, the low power factors make them unsuitable for power generation applications below 400 °C. Because both efficiency and output power are equally important, new n- and p-type materials that can work up to 400 °C are more desirable for thermoelectric power generation.

Here, we report a new Mg_2Sn -based n-type thermoelectric material that shows promise to work below 400 °C for power generation due to the narrow band gap of ~ 0.26 eV. Historically, Mg_2Sn material has been investigated less than its analogous compound Mg_2Si for thermoelectric applications due to its lower ZT (22–25). Most of the research has been focused on the alloy of Mg_2Si - Mg_2Sn with a peak ZT value of ~ 1 at 500 °C (26–28). Recently, different groups have improved the peak ZT value to 1.1–1.3 by adjusting the x value in the $\text{Mg}_2\text{Si}_{1-x}\text{Sn}_x$ solid solution (14, 29, 30). The challenges in preparing and handling these materials were the high vapor pressure and chemical activity of Mg. Methods of direct comelting with subsequent annealing, and solid-state reaction with subsequent annealing and Bridgman method were reported to synthesize Mg_2Si - Mg_2Sn alloys (22–30). Powder metallurgy route, e.g., ball milling plus hot pressing, was widely used to fabricate a variety of high-performance thermoelectric bulk materials such as Bi_2Te_3 (6, 31), PbTe (32), PbSe (33), and skutterudites CoSb_3 (16, 34). In fact, ball milling was reported to synthesize Mg_2Si and its alloys Mg_2Si - Mg_2Sn (35–38). However, the reported ZT was lower than 0.7 (27, 38), which may be due to the difficulty in avoiding oxidization of Mg. Here, we report a successful synthesis of an Sn-dominated composition $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ through ball milling and hot pressing to achieve a ZT of 1.4 at 450 °C and power factor PF of $55 \mu\text{W}\cdot\text{cm}^{-1}\cdot\text{K}^{-2}$ at 350 °C. Calculations show that these could yield a leg efficiency η of 10.5%, and output power density ω of $6.6 \text{ W}\cdot\text{cm}^{-2}$ at $T_h = 400$ °C and $T_c = 50$ °C, which will be very useful for the vast amount of waste heat sources at up to 400 °C and concentrated solar energy conversion applications.

Experiment and Characterization

Synthesis. Elemental powders, including magnesium (Mg, 99.98%; Alfa Aesar), tin (Sn, 99.8%; Alfa Aesar), and germanium (Ge, 99.999%; Alfa Aesar), were weighed according to the stoichiometric Mg_2Sn and $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$. Here, antimony (Sb, 99.99%; Alfa Aesar) and slight extra magnesium was used to adjust the carrier concentration. The element mixtures were then subjected to mechanical ball milling for up to 20 h. The ball-milled powders were then loaded into a graphite die with an inner diameter of 12.7 mm and hot pressed into bulk samples by direct current-induced hot pressing at 600–750 °C for 2 min.

Crystal Structure. X-ray diffraction (XRD) measurements were conducted on two systems: PANalytical multipurpose diffractometer with an X'celerator detector (PANalytical X'Pert Pro) and a Bruker D2 PHASER system. The lattice parameters of $\text{Mg}_2\text{Sn}_{1-x}\text{Ge}_x\text{Sb}_y$ were calculated by the Rietveld refinement method, which was done in Fullprof suite by using a cubic structure (space group: $Fm\bar{3}m$, no. 225) as the starting structure. The microstructures of the samples were studied by a JEOL 6340F scanning electron microscope (SEM) and a JEOL 2100F transmission electron microscope (TEM). The composition analysis was conducted by energy-dispersive X-ray spectroscopy (EDS) inside the SEM.

Thermoelectric Transport Properties. The electrical resistivity was measured by a reverse dc-current four-point method, whereas the Seebeck coefficient was determined by the slope of the voltage difference versus temperature difference curve based on a static temperature difference method. The simultaneous measurement of electrical resistivity and Seebeck coefficient was conducted on a commercial system (ZEM-3; ULVAC). The thermal conductivity was calculated from the relationship $\kappa = D_C\rho d$, where D , C_p , and d are the thermal diffusivity, specific heat, and volumetric density, respectively. The

thermal diffusivity was measured by the laser flash method with a commercial system (LFA457; Netzsch). The specific heat capacity was determined by a differential scanning calorimeter (DSC 404 C; Netzsch). The volumetric density was measured by the Archimedes method. The Hall coefficient, R_H , was carried out on a commercial system (PPMS; Quantum Design) with a magnetic field up to 6 T and an electrical current of 10–20 mA.

Results and Discussion

Fig. 1 shows the X-ray diffraction patterns of the ball-milled powder and hot-pressed bulk Mg_2Sn samples. The particle size of the starting materials is important to obtain phase-pure samples. Because both the major starting materials Mg and Sn are very soft, they easily stick on the wall of the stainless-steel jar. Luckily, the reacted product Mg_2Sn is brittle, which can be ball-milled into nano powders. Even though a structure transition from cubic to hexagonal was reported in Mg_2Sn by continuous mechanical ball milling (37), we observed only cubic phase in the ball-milled Mg_2Sn nano powders, as shown in Fig. 1A. It demonstrates that small stress was left in ball-milled nano powders. The hot-pressed bulk has the same cubic Mg_2Sn phase. No oxide impurity was identified, as shown Fig. 1B. To further improve the thermoelectric properties, we explored the partial substitution of Sn with Ge and found the same cubic phase was obtained in hot-pressed $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$. The Rietveld refinement shows lattice parameter a of 6.7670 Å for Mg_2Sn and 6.6786 Å for $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$, which is in good agreement with the Vegard's law for the $\text{Mg}_2\text{Sn}_{1-x}\text{Ge}_x$ system. Fig. 2 shows the microstructures of the as-fabricated $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$. The SEM image of the freshly fractured surface, as shown in Fig. 2A, shows the grain size distribution ranges from 0.5 to 2 μm . Smaller grains of ~ 300 nm were also identified in the TEM image, as shown in Fig. 2B. Furthermore, some nano inclusions of 5–10 nm were observed in the high-resolution TEM images, as shown in Fig. 2C and D. Such nano inclusions are most likely due to stress caused by the

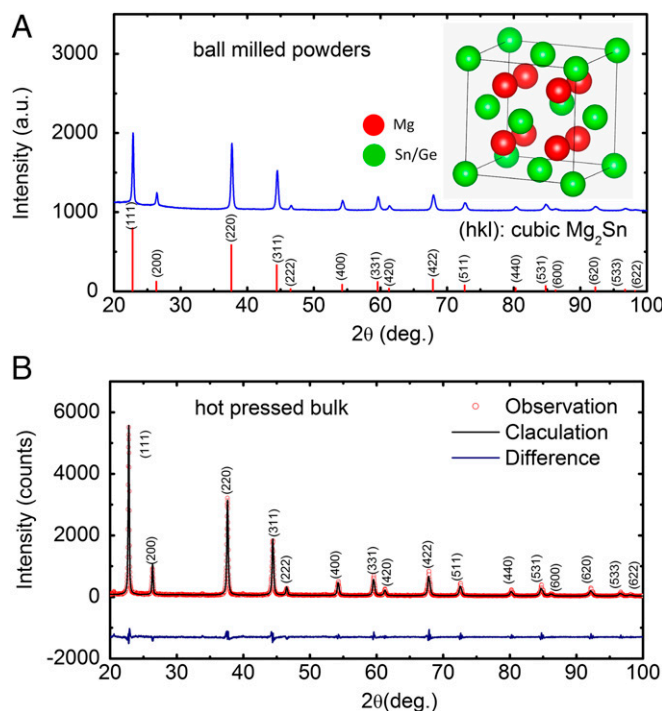


Fig. 1. X-ray diffraction patterns of $\text{Mg}_2\text{Sn}_{1-x}\text{Ge}_x$. (A) XRD pattern of ball-milled powders, which matches the standard cubic Mg_2Sn phase (PDF no. 03-065-2997), and (B) XRD pattern of the hot-pressed bulk polycrystalline Mg_2Sn with Rietveld refinement. The Inset of A is the schematic figure of cubic Mg_2X ($\text{X} = \text{Sn}$ and Ge) crystalized in the space group of $Fm\bar{3}m$ (no. 225).

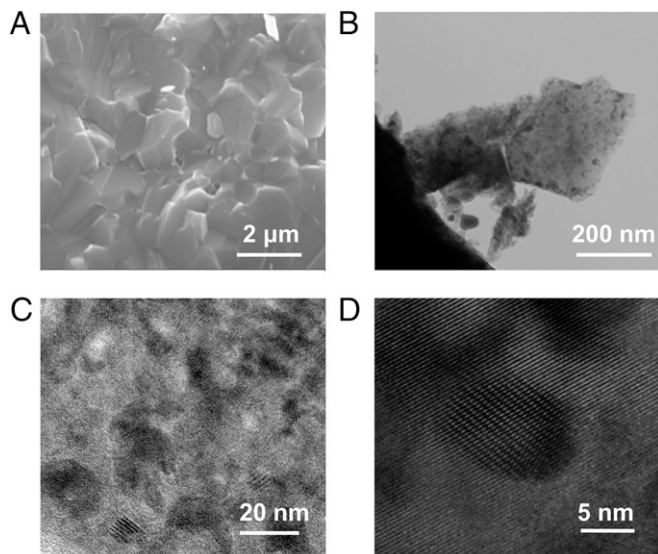


Fig. 2. Microstructures of $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ by SEM and TEM. (A) SEM image of freshly fractured surface. (B) TEM image of a single grain. (C) Nano inclusions identified by the stress contrast. (D) Nano inclusion with coherent boundary within the matrix.

slight composition fluctuation within the single crystalline grain, similar to the Ag/Sb-codoped PbTe (5) and Sn/Te-codoped CoSb₃ (34). These nano inclusions are the active phonon-scattering centers. It is noted that the observed nano inclusions have coherent boundaries within the matrix, which should have less impact on the transport of the electrons (7). Currently, we have not yet fully understood the real formation mechanism of these nano inclusions. We speculated that the possible reason would be related to the extra Mg entering the interstitial sites or slight Ge segregation. The measured compositions of $\text{Mg}_{2.06}\text{Sn}_{0.728}\text{Ge}_{0.25}\text{Sb}_{0.022}$ by EDS inside SEM are 66.6 at.% for Mg, 21.9 at.% for Sn, 9.9 at.% for Ge, and 1.6 at.% for Sb, as shown in Fig. S1, which are close to the nominal composition.

Fig. 3 shows the thermoelectric properties of the hot-pressed Mg_2Sn and $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$. Antimony is used to further tune the carrier concentration. The carrier concentrations of Mg_2Sn and $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ are $1.8 \times 10^{20} \text{ cm}^{-3}$ and $3.0 \times 10^{20} \text{ cm}^{-3}$, respectively. The net carrier of each Sb atom is estimated to be approximately one carrier per atom, which is independent of Ge content in the $\text{Mg}_2\text{Sn}_{1-x}\text{Ge}_x$ system. The electrical resistivity and Seebeck coefficient increase almost linearly below 300 °C, demonstrating degenerate semiconductor behavior, as shown in Fig. 3 A and B. The Fermi energy ($E_F/k_B T$) calculated from the Seebeck coefficient is 0.031–0.049 eV, equal to 1.2–1.9 $k_B T$. Above 300 °C, the Seebeck coefficient of Mg_2Sn tends to saturate, which results from the onset of the bipolar effect due to the narrow band gap. However, $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ still follows the linear increase above 300 °C, which suggests widening of the band gap. This is consistent with the earlier reported result: Ge continuously increases the band gap from 0.26 eV for $x = 0$ to 0.41 eV for $x = 0.4$ in $\text{Mg}_2\text{Sn}_{1-x}\text{Ge}_x$ (39). One of the advantages associated with the alloying effect of Ge is the higher power factor of $45 \mu\text{W}\cdot\text{cm}^{-1}\cdot\text{K}^{-2}$ near room temperature, as shown in Fig. 3C. This value is even higher than that of the textured $\text{Bi}_2\text{Te}_{2.7}\text{Se}_{0.3}$ polycrystalline sample ($\sim 39 \mu\text{W}\cdot\text{cm}^{-1}\cdot\text{K}^{-2}$) (40). The maximum power factor of $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ has reached $55 \mu\text{W}\cdot\text{cm}^{-1}\cdot\text{K}^{-2}$ at 350 °C. Due to the weak temperature-dependent behavior, the $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ sample also has a higher average power factor of $\sim 52 \mu\text{W}\cdot\text{cm}^{-1}\cdot\text{K}^{-2}$ over the temperature range of 25–450 °C, which is 13–16% higher than that of the best $\text{Mg}_2\text{Sn}_{1-x}\text{Si}_x$

samples [$\sim 46 \mu\text{W}\cdot\text{cm}^{-1}\cdot\text{K}^{-2}$ for $\text{Mg}_2\text{Sn}_{0.7}\text{Si}_{0.3}$ (29), $\sim 45 \mu\text{W}\cdot\text{cm}^{-1}\cdot\text{K}^{-2}$ for $\text{Mg}_2\text{Sn}_{0.6}\text{Si}_{0.4}$ (14)]. We have repeated the experiments many times and found fair reproducibility. Fig. 3D shows the power factor of three batches of $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ samples.

According to the theoretical understanding, Mg_2X ($\text{X} = \text{Si}, \text{Ge}, \text{Sn}$) are indirect semiconductors, in which the top of valence band is located at Γ point whereas the bottom of the conduction band is located at X point (39). For Mg_2Sn , the lowest conduction band is unoccupied Mg(3s) band (identified as X_H -band), followed by a hybridized Mg(3s)-Sn(5p) band (identified as X_L -band) that is slightly higher than the lowest band with band edge difference of $E_\Delta = 0.16 \text{ eV}$ (41). The positions of X_L -band and X_H -band in Mg_2Ge are just reversed (39). A composition-related band convergence was earlier identified. Similar composition-dependent band convergence was also observed in $\text{PbTe}_{1-x}\text{Se}_x$ (42) and $\text{Mg}_2\text{Sn}_{1-x}\text{Si}_x$ (29) systems. In the $\text{Mg}_2\text{Sn}_{1-x}\text{Ge}_x$ system, the contribution of the additional band (X_L -band) to the total electronic transport significantly increases with the increasing of Ge content. A direct result of the band convergence effect is the increased S^2n from $2.45 \times 10^{14} \text{ V}^2\cdot\text{K}^{-2}\cdot\text{m}^{-3}$ for $x = 0$ to $5.40 \times 10^{14} \text{ V}^2\cdot\text{K}^{-2}\cdot\text{m}^{-3}$ for $x = 0.25$ in $\text{Mg}_2\text{Sn}_{1-x}\text{Ge}_x$. The increased S^2n is confirmed by the increased carrier effective mass from $m^* = 2 m_0$ for $x = 0$ to $m^* = 3.5 m_0$ for $x = 0.25$. Here, the carrier effective mass was estimated from the measured Hall carrier concentration and the Seebeck coefficient (4):

$$S = -\frac{k_B}{e} \left(\frac{(r+5/2)F_{r+3/2}(\xi)}{(r+3/2)F_{r+1/2}(\xi)} - \xi \right), \quad [3]$$

$$n_H = \frac{2}{r_H} \left(\frac{2\pi m_e^* k_B T}{h^2} \right)^{3/2} F_{1/2}(\xi), \quad [4]$$

$$r_H = \frac{3}{4} \frac{F_{1/2}(\xi)F_{-1/2}(\xi)}{F_0^2(\xi)}, \quad [5]$$

where $F_n(\xi)$ is the Fermi integration, ξ is the Fermi energy, and r is the scattering factor in the relaxation time approximation. The increased carrier effective mass m^* could be direct evidence for the increased contribution of the additional X_L -band to the main X_H -band owing to the shrinkage of the band edge difference E_Δ .

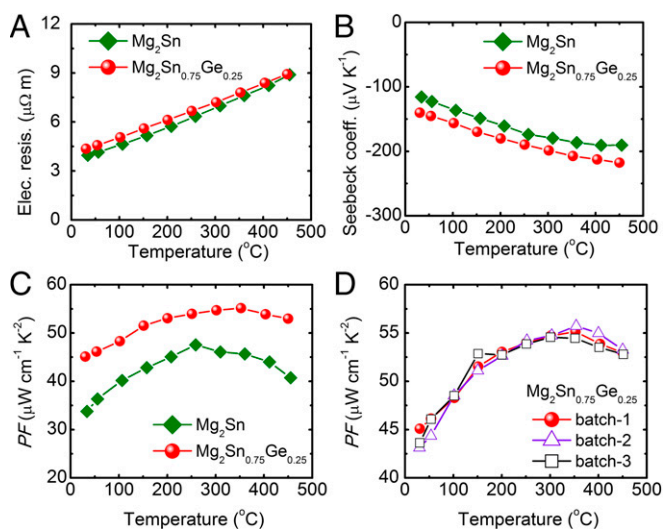


Fig. 3. Temperature- and composition-dependent electrical transport properties of $\text{Mg}_2\text{Sn}_{1-x}\text{Ge}_x$. (A) Electrical conductivity. (B) Seebeck coefficient. (C) Power factor (PF). (D) PF of $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ from different batches.

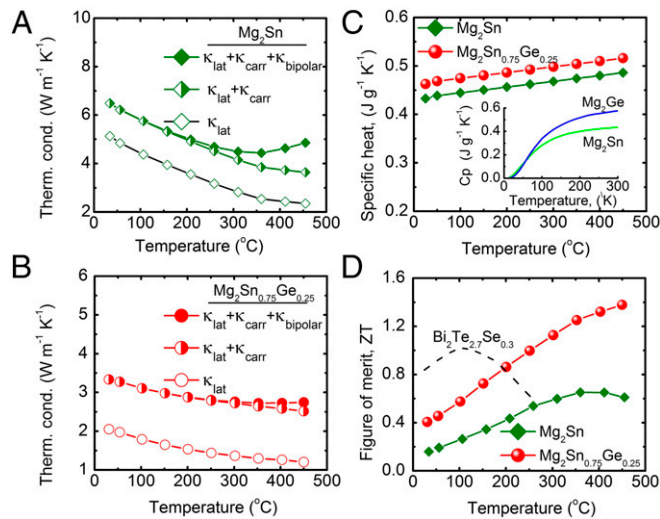


Fig. 4. Temperature-dependent thermal properties and ZT values of Mg₂Sn_{1-x}Ge_x. (A) Thermal conductivity of Mg₂Sn. (B) Thermal conductivity of Mg₂Sn_{0.75}Ge_{0.25}. (C) Specific heat of Mg₂Sn and Mg₂Sn_{0.75}Ge_{0.25}. (D) ZT values. The Inset of C is the specific heat of pure Mg₂Sn and Mg₂Ge at low temperature from 1 to 300 K (43, 44). The ZT values of Bi₂Te_{2.7}Se_{0.3} (40) are plotted in D for comparison.

However, only increased S^2n cannot guarantee the improvement of the power factor if carrier mobility is decreased too much due to the addition of Ge. In our case, a decrease in carrier mobility from 86 cm²·V⁻¹·s⁻¹ for $x = 0$ to 46 cm²·V⁻¹·s⁻¹ for $x = 0.25$ was observed, which is considered as the result of the alloying scattering. However, the ratio of decreasing μ is less than that of increasing S^2n in the Mg₂Sn_{1-x}Ge_x system, which therefore leads to a raised PF. The alloying scattering to charge carriers mainly resulted from the lattice deformation due to alloying atoms. Physically, the effect of lattice deformation on the carrier mobility is quantified through the term of deformed potential energy fluctuation U_d on the basis of the relaxation time approximation (30):

$$\mu_{al} = \frac{eh^4}{9\sqrt{2}\pi^5 x(1-x)(\kappa_B T)^{1/2}} \frac{N_0}{U_d^2 (m_b^*)^{5/2}} \frac{F_0(\xi)}{F_{1/2}(\xi)}, \quad [6]$$

where N_0 is the atomic number in unit volume; x , the fractional concentration of the alloy atom at a given lattice site; and m_b^* , the average single valley density of states effective mass, related to the effective mass through $m^* = N_V^{2/3} m_b^*$. Chemically, the lattice deformation, due to the alloying atom or impurity atom, is connected to the change of the chemical bonding strength and length. Here, we estimated the potential energy fluctuation U_d by first extracting the μ_{al} from the measured carrier mobility μ . The acoustic phonon scattering needs to be taken into account in addition to the alloying disorder scattering for electrons through the Matthiessen's rule as follows:

$$\frac{1}{\mu} = \frac{1}{\mu_{ph}} + \frac{1}{\mu_{al}}. \quad [7]$$

μ_{ph} is the carrier mobility with only acoustic phonon scattering. Here, we use the measured μ of Mg₂Sn (~ 86 cm²·V⁻¹·s⁻¹) as an approximation for the μ_{ph} in Sn-enriched Mg₂Sn_{1-x}Ge_x. By applying Eqs. 6 and 7 to the carrier mobility of Mg₂Sn_{1-x}Ge_x ($x = 0.01, 0.03, 0.05, 0.1, 0.15, 0.2, 0.25$), we got the potential energy fluctuation U_d of 0.07–0.2 eV. This value is much smaller than 0.7 eV for U_d obtained in Mg₂Sn_{0.55}Si_{0.45} (30), which suggested

that the change of the chemical bond due to Ge is smaller than that due to Si in the Mg₂Sn matrix. The most likely reason is that the atomic size of Ge is closer to Sn than that of Si.

Thermal conductivity of as-fabricated Mg₂Sn and Mg₂Sn_{0.75}Ge_{0.25} is calculated from the measured density, thermal diffusivity, and specific heat, and is shown in Fig. 4 A and B. The measured specific heat is shown in Fig. 4C with the reference data (44, 45) for comparison. At 450 °C, the specific heat of both samples is $\sim 8\%$ higher than the Dulong–Petit value. The lattice thermal conductivity was calculated by extracting both the contribution of charge carriers and bipolar effect (4):

$$\kappa = \kappa_{lat} + \kappa_{carr} + \kappa_{bipolar}, \quad [8]$$

$$\kappa_{carr} = L\sigma T, \quad [9]$$

$$\kappa_{bipolar} = \frac{\sigma_e \sigma_h}{\sigma_e + \sigma_h} (S_e - S_h)^2 T. \quad [10]$$

Here, an equivalent two-band model (one conduction band and one valence band) was used to fit the temperature-dependent electrical conductivity and Seebeck coefficient, and then extract the necessary parameters (including carrier mobility, carrier effective mass, band gap, and Fermi energy) for calculating σ_e , σ_h , S_e , and S_h , and finally the $\kappa_{bipolar}$. A significant reduction in lattice thermal conductivity from 5.13 W·m⁻¹·K⁻¹ (Mg₂Sn) to 2.27 W·m⁻¹·K⁻¹ (Mg₂Sn_{0.75}Ge_{0.25}) at 25 °C was seen due to alloying scattering to phonons. As a combination of the increased power factor and the decreased thermal conductivity, the sample Mg₂Sn_{0.75}Ge_{0.25} has a peak ZT value of 1.4 at 450 °C, which is 130% higher than that of Mg₂Sn. The ZT value of the best n-type Bi₂Te_{2.7}Se_{0.3} is presented in Fig. 4D for comparison. Clearly, Mg₂Sn_{0.75}Ge_{0.25} has higher ZT than Bi₂Te_{2.7}Se_{0.3} above 200 °C. Therefore, a segmented leg of Bi₂Te_{2.7}Se_{0.3} with Mg₂Sn_{0.75}Ge_{0.25} could result in a higher efficiency and output power density in the temperature range of 25–450 °C.

To evaluate the thermoelectric performance of Mg₂Sn_{0.75}Ge_{0.25} in terms of both the conversion efficiency and output power, we compare Mg₂Sn_{0.75}Ge_{0.25} with some of the reported best Mg₂Sn_{1-x}Si_x samples in Fig. 5. First, the power factor of Mg₂Sn_{0.75}Ge_{0.25} is almost three times higher than that of Mg₂Si_{0.7}Sn_{0.3} made by ball milling and hot pressing (38). However, most of the Mg₂Sn_{1-x}Si_x samples synthesized by melting metallurgy showed ZT higher than 1, e.g., Mg₂Sn_{0.7}Si_{0.3} showed a ZT of 1.3 near 430 °C but was almost 15% lower in power factor, as shown in Fig. 5 A and B. To exemplify the role of PF in the performance of both efficiency and output power, we took the temperature-dependent thermoelectric transport properties into account. The energy balance equation of one-dimensional heat flow is as follows (40, 45–47):

$$\frac{d}{dx} \left(\kappa(x) \frac{dT(x)}{dx} \right) + J^2 \rho(x) - JT(x) \frac{dS(x)}{dx} - \varepsilon \sigma \frac{w}{A} (T^4(x) - T_\infty^4) = 0, \quad [11]$$

where ε , σ , w , J , and T_∞ are the emissivity, the Stephan–Boltzmann constant, perimeter of a leg, current density, and the ambient temperature, respectively. The parts on the left-hand side represent the conduction heat, Joule heat, Thomson heat, and radiation heat loss, respectively. Because the temperature is a function of position, x , we assign T_h at $x = 0$ and T_c at $x = L$ (length of the leg), so that the temperature-dependent thermoelectric properties are also a function of x . To solve this differential equation, a finite difference model is developed, where a leg is divided into n nodes, and a central difference method is applied to determine the relationship of adjacent nodes. Here, a conservative temperature boundary of $T_h = 400$ °C and $T_c = 50$ °C

was used considering the potentially thermal instability of $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ above 400 °C. For simplicity, no radiation heat loss was considered here. Fig. 5D shows leg output power under different temperature gradient, i.e., $(T_h - T_c)/L$. Under the same temperature gradient of 150 °C/mm, $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ leg could have an output power density of $\sim 6.6 \text{ W}\cdot\text{cm}^{-2}$ with a leg efficiency of 10.5%, which is better than any of the reported state-of-the-art $\text{Mg}_2\text{Sn}_{1-x}\text{Si}_x$ materials, as shown in Fig. 5E. Based on the dual index of efficiency and output power, $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ seems to be the best material for power generation in the temperature range of $T_h = 400 \text{ °C}$ and $T_c = 50 \text{ °C}$. It may be arguable that the same output power could also be achieved by using larger temperature gradient for samples with smaller PF and smaller κ . However, the main issue is that a larger temperature gradient will result in a larger thermal stress at the contact interface between thermoelectric legs and electrode, which will significantly shorten the lifetime of the device. By comparing Fig. 5C and F, it is clearly shown that the material with higher power factor is preferred to achieve higher output power for the same ZT values. The figure of “average ZT versus average PF ” could be a new criterion for exploring new thermoelectric materials, or for boosting the thermoelectric performance of the known materials. Furthermore, for real applications, the price of the raw materials needs to be taken into consideration. The estimated price of $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ raw material is $\sim \$190/\text{kg}$ according to the US Geological Survey Data Series (2013) (48). It is comparable to the conventional n-type $\text{Bi}_2\text{Te}_{2.7}\text{Se}_{0.3}$ ($\$170/\text{kg}$) and p-type $\text{Bi}_{0.4}\text{Sb}_{1.6}\text{Te}$ ($\$209/\text{kg}$) but cheaper than $\text{Hf}_{0.75}\text{Zr}_{0.25}\text{Sn}_{0.99}\text{Sb}_{0.01}$ ($\$235/\text{kg}$) and In_4Se_3 ($\$580/\text{kg}$). The cost of $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ can be further reduced from $\$190/\text{kg}$ to $\$150/\text{kg}$ without reduction in ZT by partially replacing the 20% germanium with silicon.

Finally, we would like to address a few general rules for exploring novel thermoelectric compounds in terms of the criteria of efficiency (high ZT) plus effectiveness (high PF). First, choose the narrow band gap compounds with high-symmetry crystalline structure for high degenerate energy valley, and hence high PF . Mg_2X ($\text{X} = \text{Si}, \text{Ge}, \text{Sn}$) has the cubic structure with narrow band gap. Second, choose heavy atoms for low κ_{lat} . For the isoelectronic compounds Mg_2Sn and Mg_2Si , Sn is heavier than Si, so Mg_2Sn should have smaller phonon group velocity, and hence lower κ_{lat} . Third, choose the compounds with smaller electronegativity difference between anions and cations for high carrier mobility. For this rule, Mg_2Sn is comparable with Mg_2Si . Fourth, balance the effect of alloying element on decreasing κ_{lat} and μ . Most of the best thermoelectric materials are compounds with a sublattice filled by two or three isoelectronic atoms, e.g., Te and Se in n-type $\text{Bi}_2\text{Te}_{2.7}\text{Se}_{0.3}$; Bi and Sb in p-type $\text{Bi}_{0.4}\text{Sb}_{1.6}\text{Te}_3$; and Hf, Zr, and Ti in n-type $(\text{Hf}, \text{Zr}, \text{Ti})\text{NiSn}$. Conventionally, Sn is added to Mg_2Si to optimize the ZT value, whereas Ge is used in Mg_2Sn in our case. The atomic size of Ge is closer to Sn than that of Si, and therefore it has less impact on the carrier mobility, so it should be good for high power factor in Mg_2Sn . Fifth, use the compositional band-crossing effect to optimize the weight mobility $\mu(m^*)^{3/2}$, and hence achieve high PF . This is a dominant effect for $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ to have both high PF and ZT . Sixth, apply the nano approaches to selectively scatter the phonons rather than electrons. This effect is also an important factor in our case. Getting to technologically useful materials in thermoelectric power generation could be a slow process, but these guiding rules will be helpful in exploring new promising thermoelectric materials.

Conclusions

We have successfully synthesized Mg_2Sn -based materials by ball milling and hot pressing. The composition of $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ shows an average ZT of 0.9 and an average PF of $52 \mu\text{W}\cdot\text{cm}^{-1}\cdot\text{K}^{-2}$ over the temperature range of 25–450 °C, and also a peak ZT of 1.4

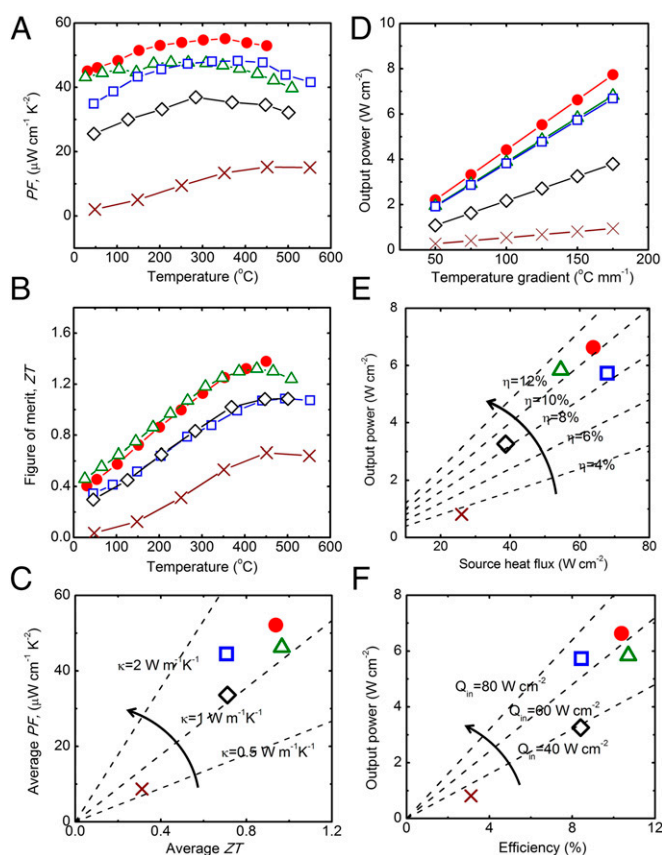


Fig. 5. Comparison between $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ with the state-of-the-art $\text{Mg}_2\text{Sn}_{1-x}\text{Si}_x$. (A) Temperature-dependent power factor (PF). (B) Temperature-dependent ZT . (C) Average PF versus average ZT . (D) Leg output power density as function of temperature gradient. (E) Leg output power density as a function of source heat flux. (F) Output power versus efficiency. Samples made by ball milling and hot pressing route: filled circles (●) are $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ in this work, and crosses (×) are $\text{Mg}_2\text{Sn}_{0.3}\text{Si}_{0.7}$ from ref. 38. Samples made from melting route: unfilled squares (□) are $\text{Mg}_2\text{Sn}_{0.6}\text{Si}_{0.4}$ from ref. 14, unfilled diamonds (◇) are $\text{Mg}_2\text{Sn}_{0.6}\text{Si}_{0.4}$ from ref. 28, and unfilled triangles (△) are $\text{Mg}_2\text{Sn}_{0.7}\text{Si}_{0.3}$ from ref. 29. The slope of the dashed line in C is the κT_{avg} with $\kappa = 0.5, 1$, and $2 \text{ W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$ and $T_{\text{avg}} = 225 \text{ °C}$. The slope of the dashed line in E is the efficiency η of 4%, 6%, 8%, 10%, and 12%. The dashed line in F is heat flux at hot side Q_{in} of 40, 60, and 80 $\text{W}\cdot\text{cm}^{-2}$.

at 450 °C and peak PF of $55 \mu\text{W}\cdot\text{cm}^{-1}\cdot\text{K}^{-2}$ at 350 °C. Theoretically, $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$ leg could have an efficiency η of 10.5%, and an output power density ω of $6.6 \text{ W}\cdot\text{cm}^{-2}$ under a temperature gradient of $150 \text{ °C}\cdot\text{mm}^{-1}$ for $T_h = 400 \text{ °C}$ and $T_c = 50 \text{ °C}$. Compared with the reported $\text{Mg}_2\text{Sn}_{1-x}\text{Si}_x$ system, the higher power factor obtained in $\text{Mg}_2\text{Sn}_{1-x}\text{Ge}_x$ is associated with the increased S^2n attributable to the band convergence and higher carrier mobility, which in turn are due to the smaller size difference between Sn and Ge than that between Sn and Si. Furthermore, coherent nano inclusions were identified in the $\text{Mg}_2\text{Sn}_{1-x}\text{Ge}_x$ materials. Recently, 5–10% Ge-doped $\text{Mg}_2\text{Si}_{1-x}\text{Sn}_x$ have also been reported (49, 50); however, the reported power factor $PF_{\text{max}} = 35 \mu\text{W}\cdot\text{cm}^{-1}\cdot\text{K}^{-2}$ (ref. 50) is still much lower than what we are reporting in this paper ($PF_{\text{max}} = 55 \mu\text{W}\cdot\text{cm}^{-1}\cdot\text{K}^{-2}$ in $\text{Mg}_2\text{Sn}_{0.75}\text{Ge}_{0.25}$).

ACKNOWLEDGMENTS. This work is partially supported by Solid-State Solar-Thermal Energy Conversion Center, an Energy Frontier Research Center funded by the US Department of Energy, Office of Science, Office of Basic Energy Science under Award DE-SC0001299/DE-FG02-09ER46577 (materials synthesis and characterizations), and partially by Concentrated Solar

Thermoelectric Power (CSP), a Department of Energy SunShot CSP grant, under Award DE-EE0005806 (leg efficiency and output power density calculation). This work is also supported in part by US Air Force Office of Scientific Research Grant FA9550-09-1-0656, the T. L. L. Temple Foundation,

the John J. and Rebecca Moores Endowment, and the State of Texas through the Texas Center for Superconductivity at the University of Houston. B.L. also acknowledges the New Faculty Award by University of Houston.

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